Design and Develop a Control System for Six Degree of Freedom Soft Actuator

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Introduction

- Soft robotics technology, especially in medical applications, promises to revolutionize the way doctors, surgeons, and healthcare providers interact with patients.
- Soft robotic systems offer inherent safety and adaptability, making them particularly suited for delicate interactions in diverse medical scenarios, from rehabilitation devices to surgical tools [1].
- The introduction of 6 - Degree of Freedom (DOF) in soft robotics elevates the precision and versatility of robotic movements to mimic complex human gestures. This capability is crucial in surgeries and therapies where multi-directional manipulation is necessary within confined spaces.
- Traditional rigid robotic systems often fall short in these aspects due to their bulk and limited flexibility, which can lead to inefficiencies and higher risks in sensitive environments [2].
- The aim of this project is to develop an intuitive, user-friendly control system for a 6-DOF soft actuator, leveraging Xbox joystick inputs to enhance precision and adaptability in medical applications, thereby improving the efficacy and safety of robotic-assisted surgeries and rehabilitation procedures.

Methodology

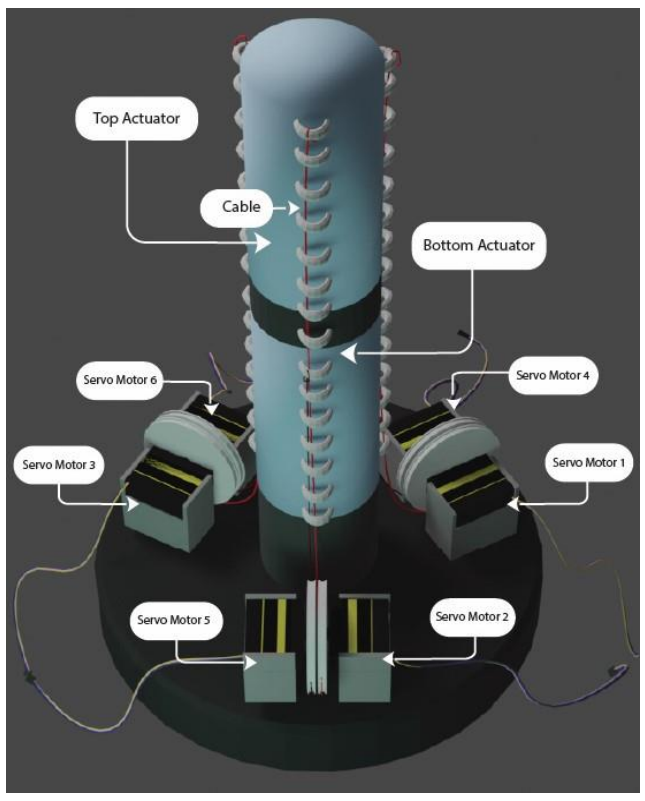

Figure 1: 3D Design of 6-DOF Soft Actuator

- The system features two vertically aligned soft actuators, identified as the Top and Bottom Actuators. These components allow for exceptional flexibility and adaptability in movement.
- It incorporates six servo motors, with three dedicated to controlling the Top Actuator and the remaining three assigned to the Bottom Actuator.
- These servo motors finely control the movement of cables attached to the actuators, enabling precise manipulation of their positioning and bending.

Figure 2: Control System Architecture

- The diagram illustrates the control system architecture that begins with inputs from an Xbox controller interfaced with MATLAB Simulink. The signals are processed and transmitted via RS232 to an Arduino controller, which is powered by an external power supply. The Arduino then controls the movements of servo motors, which operate cables to precisely adjust the position and bending of soft actuators in the system.

- **Dependent Control**
  This method controls three servo motors using a single joystick input, enabling them to reach desired positions. An additional joystick input is required to simultaneously control the remaining three motors.
- **Independent Control**
  This method employs a single joystick input to control all six servo motors, effectively avoiding the effects of cable interference.

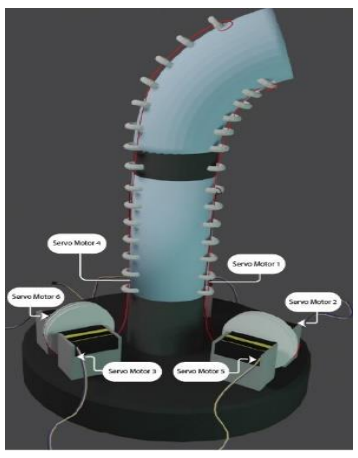

Figure 3: Top Actuator +X direction control

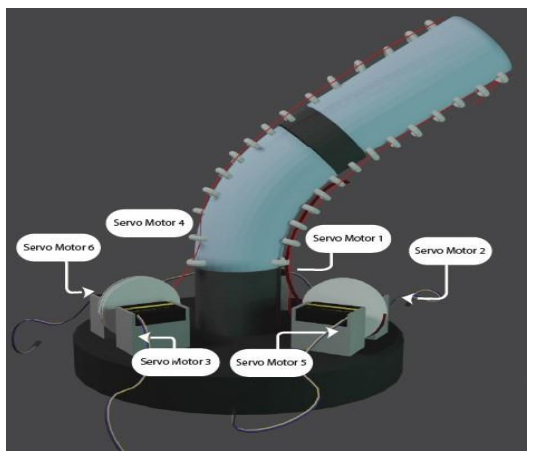

Figure 4: Bottom Actuator +X direction control

Results

- Graphs plotted in MATLAB Simulink illustrate joystick input signals and Arduino feedback signals for servo angles.
- Figure 5 depicts the control graph for the top actuator's X-direction under dependent control. When the left joystick's X-axis reaches 180°, servo motor 1 adjusts to 45°, servo motor 2 extends to 180°, and servo motor 3 retracts to 0°, facilitating movement in the -X direction. Conversely, the control mechanism reverses for +X direction movement.
- Figure 6 presents the top actuator's X-direction control graph for independent control. Here, a single joystick input prompts all servo motors to adjust simultaneously, enabling the required movement and compensating for cables interference.

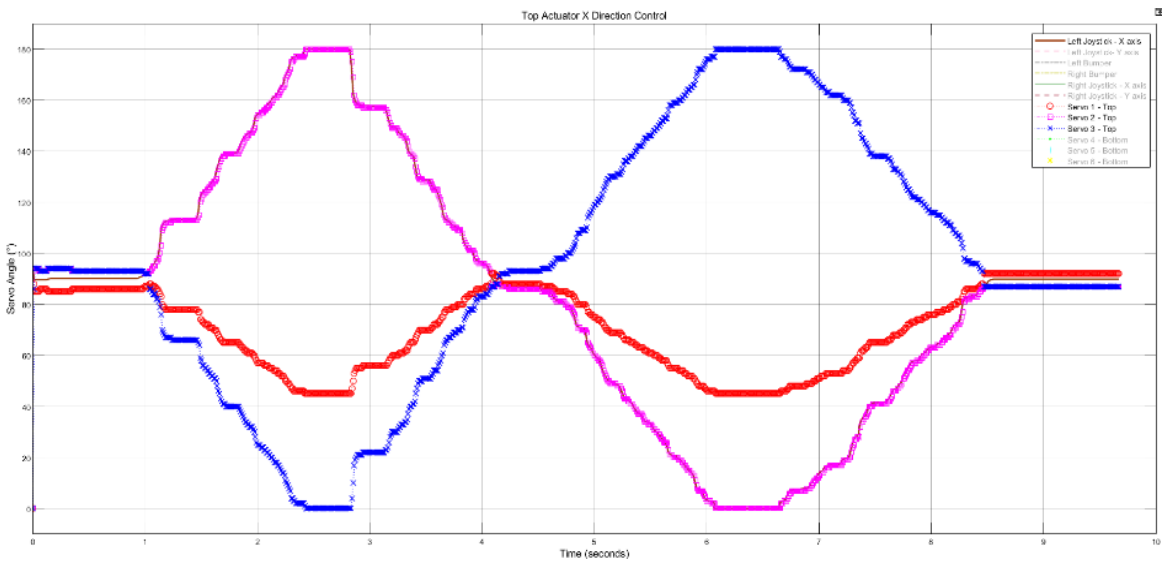

Figure 5: Example graph of dependent control

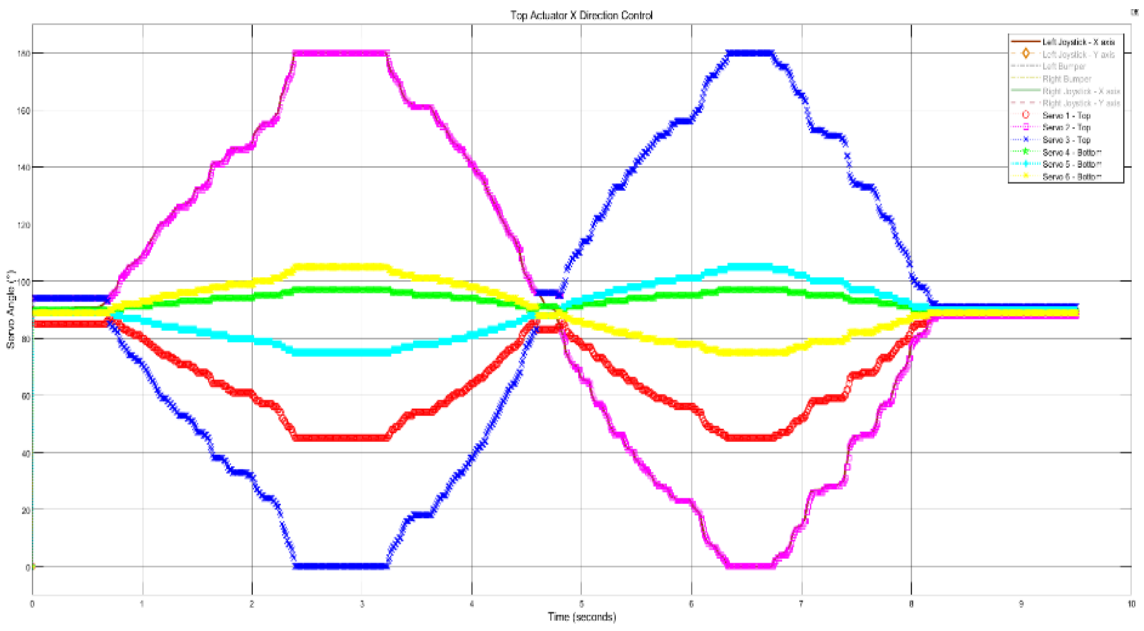

Figure 6: Example graph of independent control

Discussion and Conclusions

- ✓ Control System Performance: Evaluates how joystick inputs enhance control precision and user interface intuitiveness.
- ✓ Cable Interference Solutions: Discusses the effectiveness of independent control in mitigating cable interference issues.
- ✓ Project Contributions: Highlights the project's advancements in soft robotics control systems, with a focus on potential applications in medical robotics and other precision-dependent environments.

References

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